



Exam Prep Edition



AIR, WATER & WEATHER



Class Notes: Science / EVS

Student: [A+, Topper]

Mission: Master the Elements!

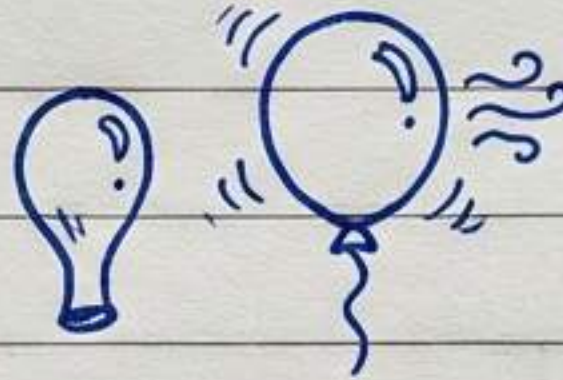


1. AIR (The Invisible Ocean)

- Air is all around us, but we cannot see it. We can only feel it.

Properties of Air:

- Takes up space
- Has weight.

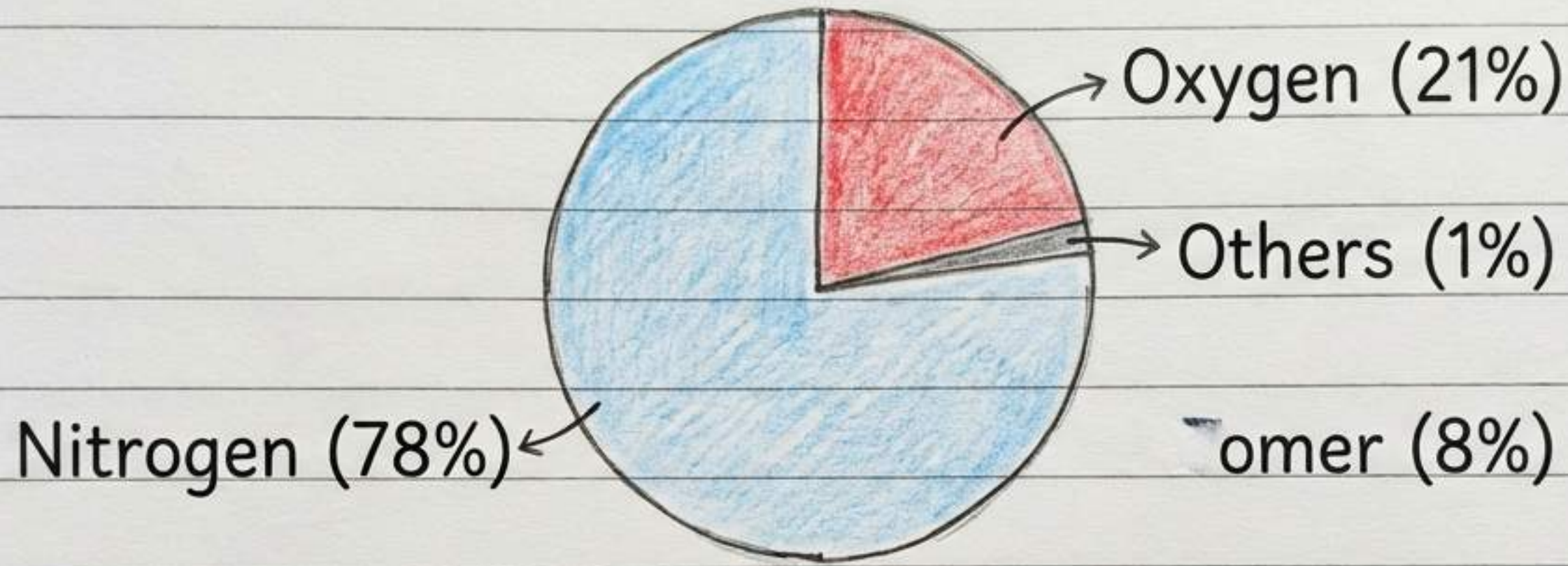


Movement Types:

- **Wind:** Moving air.
- **Breeze:** Gently moving wind (Pleasant).
- **Storm:** Fast moving wind (Dangerous/Strong).

Note: Hava aapan pahu shakat nahi, phakta anubhavu shakto. (We cannot see air, only feel it).

Composition of Air



Also contains water vapour, dust, smoke, and germs!

- Nitrogen (78%): Required for the growth of plants.
- Oxygen (21%): Essential for breathing and burning.
- Carbon Dioxide (0.03%): Released by humans/animals during breathing. Used by plants to make food (Photosynthesis).



Exam Tip: Remember: Plants need Nitrogen to grow, but CO_2 to eat!

Uses of Air

- Breathing: All living things need **Oxygen**.



- **Burning**: Fire cannot exist without air.



- Flying: Birds, airplanes, and helicopters need air to fly.



- Sailing: Wind moves sailboats.



- Energy: **Windmills** spin to create power.



- Drying: Clothes dry faster in a gentle breeze.



Trick: W.I.N.D. →
Work (Windmills),
Inhale (Breathe),
Navigate (Sail),
Dry (Clothes).

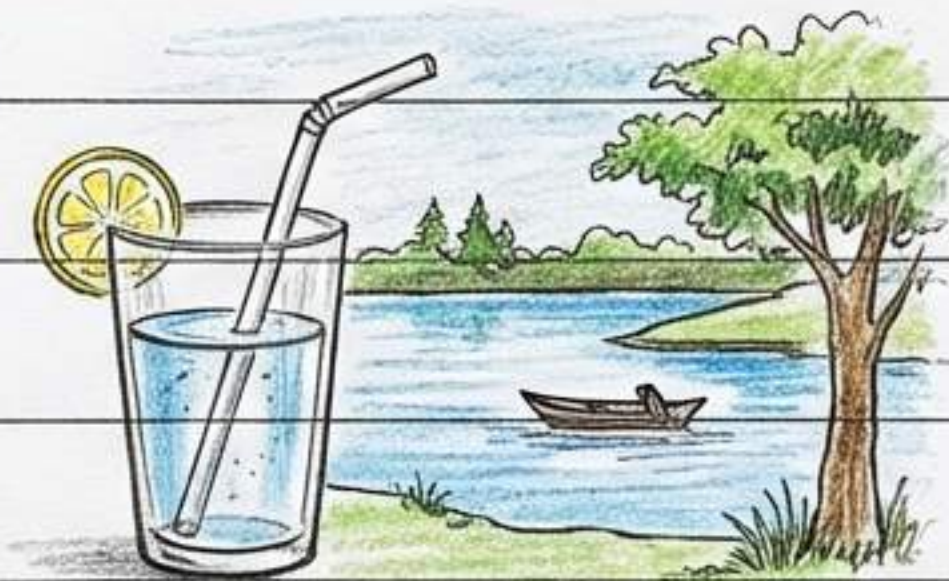
2. WATER (Our Life Source)

Fact: About 70% of Earth's surface is covered with water.

The 3 States of Water:



1. Solid: Ice / Snow
(Mountains)



2. Liquid: Water
(Rivers, Lakes)



3. Gas: Water Vapour
(Air)

Note: Paani teen avasthet aadhalte: Sthayu (Solid), Drav (Liquid),
ani Vayu (Gas).

The Water Cycle: Going UP



Step 1: Evaporation

- The Sun heats water in seas/oceans.
- Water turns into Water Vapour (Gas) and rises.

Step 2: Condensation

- Water vapour rises high and cools down.
- Turns back into tiny water droplets to form CLOUDS.

Activity Corner

Cold bottle = Wet surface?
That's Condensation!

Marathi: Evaporation = Baashpibhavan

The Water Cycle: Coming DOWN



Step 3: Precipitation

- Clouds get too heavy with water droplets.
- Water falls down as Rain, hail, or snow.

Step 4: Collection

- Rainwater flows into rivers, lakes, and oceans.
- The cycle begins again!

Trick: E.C.P.C. → Every Cloud Pours Cold water. (Evap, Cond, Precip, Collect).

Water Conservation (Don't be Hrithik!)



Who is Hrithik?

A boy who wastes water
(leaving taps open, long showers,
playing with hoses).

Why is this bad?

Water is precious. We must
save it for the future.

Good Habits (Exam Checklist):

- ☒ Close tap while brushing.
- ☒ Use a bucket instead of a hose/shower.
- ☒ Fix leaking taps immediately.

Note: Paani wachva, paani heach jivan ahe.
(Save water, water is life).

3. WEATHER

Definition: The condition of air at a specific time
(Hot, Cold, Windy, Dry).

Note: Weather changes day to day or even hour to hour! (Moment to day or growing ink bims
.. Temporary)

Sunny Days

- Sky is clear (No clouds).
- Sun shines brightly.
- Afternoons are hottest (Sun is overhead).
- Mornings/Evenings are cooler.



Windy & Cloudy Days

Windy Days

- Winds blow fast.
- Weather feels cooler.
- ★ Warning: Strong winds can cause storms (thunder/lightning).



Cloudy Days

- Sun is hidden by clouds.
- Weather is pleasant and cool.
- Might lead to rain.



Common Mistake: Don't confuse 'Windy' with 'Stormy.' A storm destroys things; wind just moves them.

4. SEASONS (The Long Cycle)

Definition: When the same weather continues for a long time (months), it is called a Season.

Season 1: SUMMER

- The hottest season.
- Days are long, nights are short.
- What we do:
 - Wear cotton clothes (absorb sweat).
 - Drink plenty of water/juices.
 - Use fans/ACs. hydrated.

Note: Havaman (Weather) badalat rahte, pan Ritu (Season) mahine tikto.

Season 2: MONSOON (Rainy)

- It rains heavily on most days.
- Accompanied by strong winds, Thunder, and Lightning.
- Dark clouds fill the sky.
- Survival Gear: Raincoats, Umbrellas, Gumboots (To stay dry).



Season 3: WINTER

- The coldest season.
- Cold, Dry winds blow.
- Mornings/Evenings are misty (foggy).
- What we do:
 - Wear woollen clothes (Sweaters, caps) to keep warm.
 - Drink hot soups/tea.
 - Use room heaters or light fires.



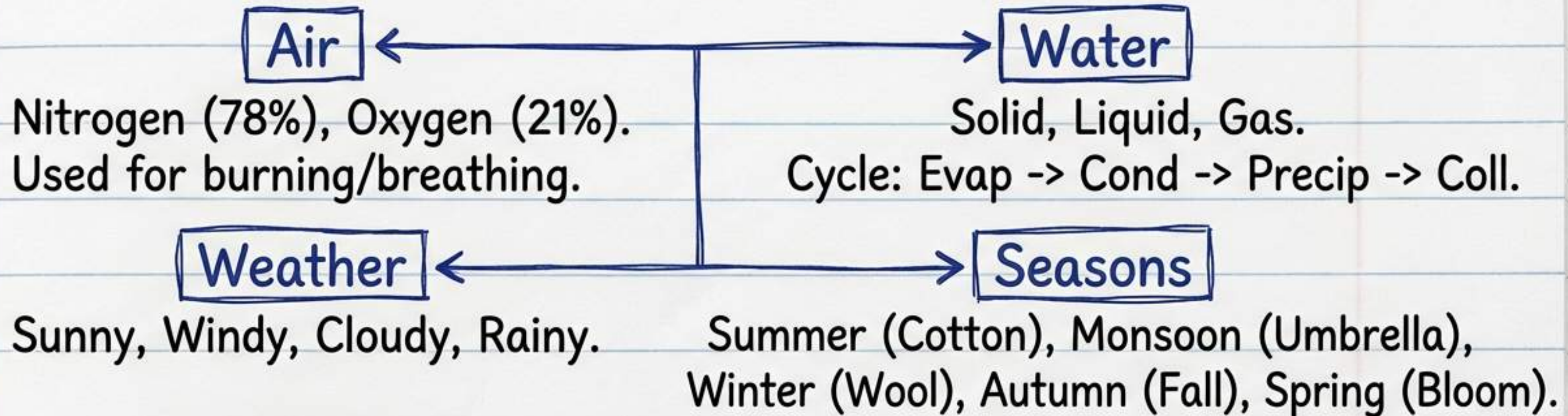
Exam Q: Why do we drink hot soup?

To raise our body temperature against the cold!

Short Seasons: Autumn vs. Spring

AUTUMN	SPRING (Basant)
- Comes before Winter. - Weather is mild (dry and windy). - Key Event: Trees shed their leaves. 	- Comes after Winter. - Weather is pleasant. - Key Event: New flowers bloom and plants grow. 

QUICK RECAP (Cheat Sheet)



Important Terms

- **Condensation:** Gas → Liquid (Clouds).
- **Evaporation:** Liquid → Gas (Steam).
- **Humid:** Water vapour in air (Monsoon).

Good Luck!

Nisargacha samtol rakhne garjeche ahe (Nature's balance is essential).